

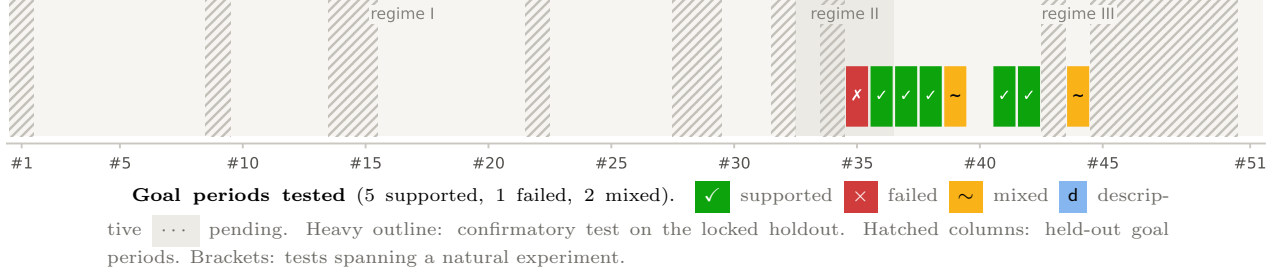
H108: Goldstone wandering: the spontaneous room direction should drift, while a fielded one stays pinned

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 1/5

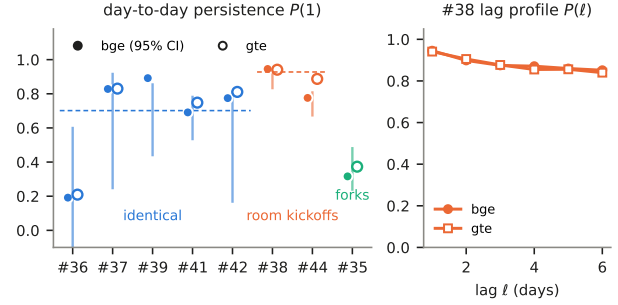
Bottom line. The room content direction persists from day to day inside a week and persists most under room-specific instructions (pooled cosine 0.93 vs 0.70), but the contrast rests on one long week and cannot yet be told apart from slower drift of a larger split.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH338 says that a spontaneously broken continuous symmetry has a soft direction. The room content difference of identical-kickoff weeks should then wander from day to day (a Goldstone mode), while room-specific instructions pin it. H108 measures the day-to-day persistence of the #best/#rest content direction in 8 non-holdout periods, corrected for sampling noise by joint relabels. **The pre-registered contrast passes, but it is fragile.** Directions persist inside a week (cosine 0.69–0.95 in 6/8 periods). The fielded weeks are more persistent (pooled 0.93 vs 0.70; ratio of decorrelation rates 4.7). The contrast rests on the one long fielded week, #38, and it is confounded with split size. The fork week does not pin.



Observables. Left: noise-corrected day-to-day persistence $P(1)$ per period (filled bge with agent-bootstrap 95% CI, open gte); dashed lines are the pooled group values. Right: #38's lag profile falls to a plateau near 0.85, the pinned signature. #44, the other fielded week, sits with the identical weeks; the fork week #35 has the least memory.

PHYSICAL MODEL

Model 11 (soft-spin $O(n)$, $n = 32$). The room order parameter $\mathbf{m}(d) = \Delta(d)$ is the daily #best – #rest content difference (style-residualized, day-centred, agent constants removed). Without a field its direction $\hat{\mathbf{m}}$ diffuses on the sphere with $D \propto 1/(N_{\text{eff}}|\mathbf{m}|^2)$; a field $\mathbf{h}$ adds a restoring torque, so the angle is an Ornstein–Uhlenbeck process with a plateau. C and E are excess cross-products over joint relabels; $P(\ell)$ estimates the lag- ℓ cosine.

$$d\hat{\mathbf{m}} = -\Gamma\chi(\mathbf{h} \cdot \hat{\mathbf{m}}_{\perp}) dt + \sqrt{2D} d\boldsymbol{\xi}_{\perp}, \quad D \simeq \sigma^2/(2N_{\text{eff}}|\mathbf{m}|^2)$$

$$P(\ell) = \frac{\sum_d C(d, d+\ell)}{\sqrt{\sum_d E(d) \sum_d E(d+\ell)}}, \quad D_{\theta} = -\ln P(1)$$

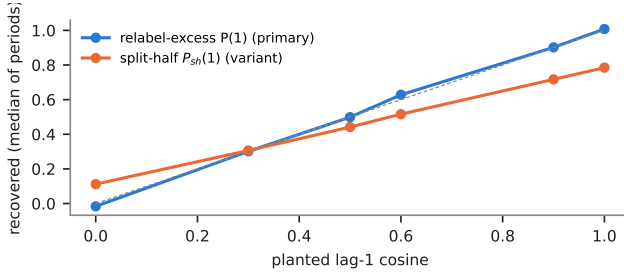
$$R_D = D_{\theta}^{\text{identical}}/D_{\theta}^{\text{fielded}}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, the useful fact is modest: inside a week the content difference between rooms keeps most of its direction (cosine about 0.7–0.9 a day), and one set of different room instructions held it nearly fixed for three weeks. Whether that is the instruction pinning the direction or simply a bigger split drifting more slowly (Goldstone predicts both) is not decided by seven periods. A work assignment alone (the forks) did not hold the content direction. Treat this as a description of room dynamics, not a lever.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 synthetic: bias, recovery, contrast power	pinned $P(1)$ 0.97–1.08; planted cosines ± 0.05 ; contrast 100%/0%/0%	passed
P1 $R_D \geq 2$, CI > 1, both models	4.7 [2.1, 5.9]; gte 5.2 [2.1, 6.4]	supported
robustness (post hoc)	without #38: 1.4 (gte 3.0); field vs size not separable	fragile
P2 D_θ rises with $1/(N \mathbf{m} ^2)$	Spearman 0.10 (bge, 5 pts), 0.40 (gte)	weak
P3 persistence beats relabel $\geq 5/8$	6/8 (gte 5/8)	met
P4 #38 plateau	slope $-0.019/\text{day}$, $P(6)$ 0.85	met
P5 forks pin	$P(1)$ 0.32; ω 0.68 vs 0.27	failed
HH338	identical rotation $\geq 2\times$ fielded	supported, fragile



Second figure. Synthetic validation on the real statement skeletons (40 worlds per setting): the relabel-excess estimator recovers the planted day-to-day cosine from 0 to 1; the within-day split-half correction named in the HH is biased low under pinning (it keeps each agent’s day deviation as signal), so it is reported but not scored.

GOAL PERIODS IN DETAIL

G35 $\times$ failed
G36 $\checkmark$ supported
G37 $\checkmark$ supported
G38 $\checkmark$ supported
G39 $\sim$ mixed
G41 $\checkmark$ supported
G42 $\checkmark$ supported
G44 $\sim$ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Two fielded periods, one of them 17 days long; #44 alone gives R_D 1.4 in bge. Goldstone diffusion slows with split size, and the fielded rooms split most; a log-log fit over seven periods leaves the field coefficient at -0.17 ± 1.33 . Identical periods have 2–4 day pairs, and agent-bootstrap CIs sit below the point estimate where duplicated agents shift the null. Day 1 is a transient (H107) and enters the pairs. The leave-one-out and size checks are post hoc.

ROUND 2

Compare fielded and identical weeks at matched split size; drop the kickoff day; test transverse kicks (does one room’s operator message rotate the direction more without a field?).

- **Essence.** whether a room’s content order is pinned by its instructions or diffuses freely, at matched order-parameter size.
- **H108-R1** Match on split size.
- **H108-R2** Exclude the kickoff-day transient.
- **H108-R3** Transverse kicks.
- **H108-R4** Why the forks do not pin.

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